

# Project Design Phase-II

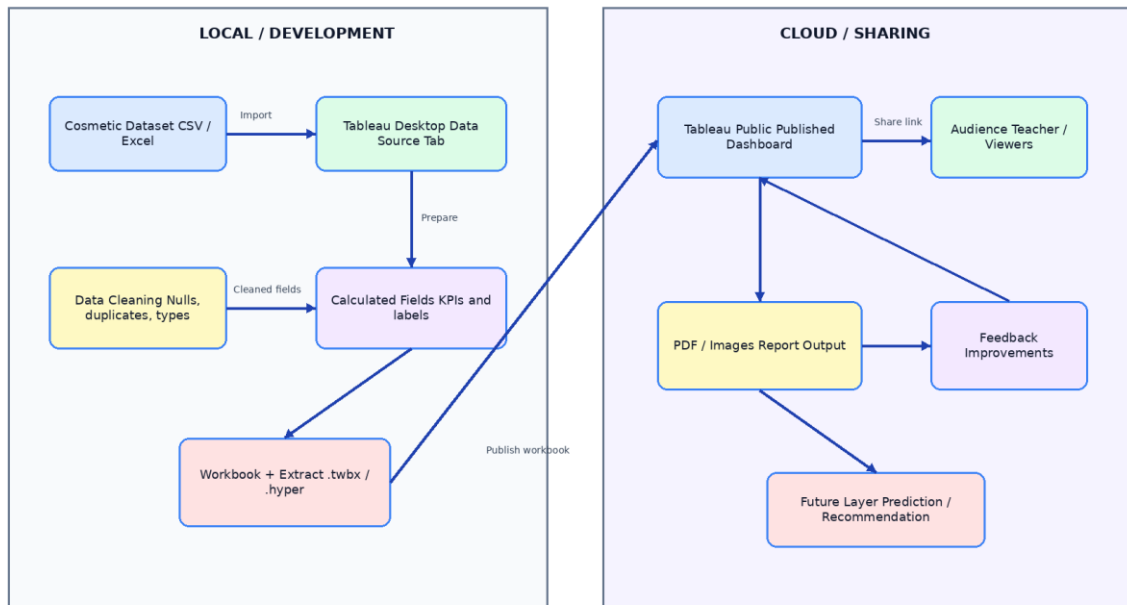
## Technology Stack (Architecture & Stack)

|               |                                     |
|---------------|-------------------------------------|
| Date          | 04 July 2026                        |
| Team ID       |                                     |
| Project Name  | Cosmetic Product Analysis Dashboard |
| Maximum Marks | 4 Marks                             |

### Technical Architecture

The deliverable includes the architecture diagram and technology tables for a Tableau-based cosmetic product analysis dashboard.

#### Solution Architecture - Cosmetic Product Analysis Dashboard



**Table-1: Components & Technologies**

| S.No | Component           | Description                                                           | Technology                                       |
|------|---------------------|-----------------------------------------------------------------------|--------------------------------------------------|
| 1    | User Interface      | Dashboard user interacts with filters, charts, KPIs and story points. | Tableau Dashboard / Tableau Public               |
| 2    | Application Logic-1 | Data loading and transformation process.                              | Tableau Desktop Data Source / Tableau Prep steps |
| 3    | Application Logic-2 | Calculated fields for KPIs and grouped analysis.                      | Tableau Calculated Fields                        |
| 4    | Application Logic-3 | Visualization and dashboard logic.                                    | Tableau Worksheets, Dashboard Actions, Filters   |
| 5    | Database            | Source file storing cosmetic product attributes.                      | CSV / Excel / Local file                         |
| 6    | Cloud Database      | Online sharing layer for workbook/dashboards.                         | Tableau Public cloud hosting                     |
| 7    | File Storage        | Workbook, extract, screenshots and report files.                      | Local filesystem, .twbx, .hyper, PDF             |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                          | Technology                                       |
|------|--------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------|
| 1    | Open-Source Frameworks   | Base dashboard does not require coding; optional report charts can use Python.       | Python, Pandas, Matplotlib (optional)            |
| 2    | Security Implementations | Use non-sensitive dataset only; avoid personal data; share public version carefully. | Tableau account controls, local file permissions |
| 3    | Scalable Architecture    | Data source, extract and dashboard layers can scale independently.                   | Tableau extract architecture                     |
| 4    | Availability             | Dashboard available through published link or PDF output.                            | Tableau Public / PDF export                      |
| 5    | Performance              | Extracts and optimized filters improve load time.                                    | Tableau Hyper Extract, filtered sheets           |